

Discrete Global Grid Hexagonal Orientation Dynamics and Thermodynamic Flux Invariance in Aperture-7 Topologies

Pascal Ranoroarijaona¹

Web of Life Research Collective¹

¹Gaia Open Project / Web of Life Simulation Framework

<https://github.com/pascalranoroarijaona/WebOfLife>

Sprint 091 Preprint — October 2023

Abstract

Multiscale biogeochemical Earth system models require equal-area discrete global spatial partitions that eliminate metric singularities and edge distortions. The Uber H3 Discrete Global Grid System (DGGS) partitions the sphere via recursive Aperture-7 hexagonal subdivision over an icosahedral projection. In an Aperture-7 hierarchy, each successive resolution level scales cell area by a factor of $1/7$, inducing an alternating rotational displacement of $\alpha = \arcsin\left(\frac{\sqrt{3}}{2\sqrt{7}}\right) \approx 19.106263^\circ$ relative to the base icosahedral coordinate axes. We present the mathematical formalization and software implementation of `getApertureClassForResolution` within the *Web of Life* computational ecology engine. By partitioning resolutions into Class II (even, unrotated) and Class III (odd, rotated) symmetries, directional edge-normal vectors $\mathbf{n}_k(r)$ are rigorously mapped via $SO(2)$ transformations. We demonstrate that accounting for this orientation alternation is necessary to preserve boundary flux invariance, eliminate artificial advective divergence ($\nabla \cdot \mathbf{J}_{\text{err}} \neq 0$), and prevent unphysical numerical entropy destruction in non-equilibrium thermodynamic monads.

1 Mathematical & Geometric Formulation

1.1 Aperture-7 Partitioning and Area Scaling

Let $r \in \mathbb{N}_0 = \{0, 1, 2, \dots, 15\}$ denote the discrete resolution index. The nominal surface area $A(r)$ of a hexagonal cell at resolution r scales recursively as:

$$A(r) = \frac{A(0)}{7^r} \quad (1)$$

where $A(0)$ is the base cell area on the spherical icosahedron. The corresponding characteristic hexagonal edge length $L(r)$ is given by:

$$L(r) = \sqrt{\frac{2A(r)}{3\sqrt{3}}} = \left(\frac{1}{\sqrt{7}}\right)^r L(0) \quad (2)$$

1.2 Alternating Symmetry Classes and the $SO(2)$ Rotation Angle

Unlike Aperture-3 or Aperture-4 tilings, the centroids of child hexagons in an Aperture-7 tiling are offset from parent hexagonal symmetry axes. To achieve geometric packing, the orientation of cell vertices alternates between two distinct symmetry classes:

$$\text{ApertureClass}(r) = \begin{cases} \text{CLASS_II}, & \text{if } r \equiv 0 \pmod{2} \\ \text{CLASS_III}, & \text{if } r \equiv 1 \pmod{2} \end{cases} \quad (3)$$

The rotation angle α distinguishing Class III from Class II is derived analytically from the centroid displacement triangle across icosahedral face coordinate frames:

$$\alpha = \arcsin\left(\frac{\sqrt{3}}{2\sqrt{7}}\right) = \arctan\left(\frac{\sqrt{3}}{5}\right) \approx 0.333473172 \text{ rad} \approx 19.1062629^\circ \quad (4)$$

For a hexagonal cell at resolution r , let $k \in \{0, 1, 2, 3, 4, 5\}$ index its six faces. In the local tangent plane $\mathbb{R}^2$, the outward unit normal vector $\mathbf{n}_k(r)$ is computed via the standard rotation matrix $\mathbf{R}(\theta_r) \in \text{SO}(2)$:

$$\mathbf{n}_k(r) = \mathbf{R}(\theta_r)\mathbf{n}_k^{(0)} \quad (5)$$

where:

$$\mathbf{n}_k^{(0)} = \begin{pmatrix} \cos\left(\frac{k\pi}{3}\right) \\ \sin\left(\frac{k\pi}{3}\right) \end{pmatrix}, \quad \mathbf{R}(\theta_r) = \begin{pmatrix} \cos\theta_r & -\sin\theta_r \\ \sin\theta_r & \cos\theta_r \end{pmatrix} \quad (6)$$

with the discrete phase angle θ_r determined by:

$$\theta_r = \begin{cases} 0, & \text{for Class II } (r \text{ even}) \\ \alpha, & \text{for Class III } (r \text{ odd}) \end{cases} \quad (7)$$

2 Thermodynamic Flux Invariance Across Rotated Interfaces

2.1 Conserved State Vector and Interface Fluxes

In the *Web of Life* ecosystem simulator, each spatial cell i hosts an extensive state vector $\mathbf{S}_i \in \mathbb{R}^6$ representing stocks of conserved chemical species and thermal energy:

$$\mathbf{S}_i = [M_{C,i} \quad M_{H_2O,i} \quad M_{O_2,i} \quad M_{N,i} \quad M_{min,i} \quad U_i]^T \quad (8)$$

representing carbon [kg], water [kg], oxygen [kg], reactive nitrogen [kg], bioavailable minerals [kg], and internal thermal energy [kJ].

Across the interface $\partial\Omega_{i,k}$ between cell i and its k -th neighbor $j = \mathcal{N}_k(i)$, the combined advective and diffusive flux vector of species s is:

$$\mathbf{j}_s = \mathbf{u}\rho_s - D_s\nabla\rho_s \quad (9)$$

where $\mathbf{u} \in \mathbb{R}^2$ is fluid/wind velocity, ρ_s is volumetric mass density, and D_s is the diffusion coefficient. The total discrete mass flow $\Phi_{s,k}$ transferred across edge k of height h is:

$$\Phi_{s,k} = (\mathbf{j}_s \cdot \mathbf{n}_k(r)) L(r)h \quad (10)$$

The discrete-time update over interval Δt enforces anti-symmetry:

$$\Delta M_{s,i \rightarrow j} = \Phi_{s,k}\Delta t, \quad \Delta M_{s,j \rightarrow i} = -\Delta M_{s,i \rightarrow j} \quad (11)$$

ensuring strict global mass conservation ($\sum_i \Delta M_{s,i} = 0$).

2.2 Numerical Divergence and Second Law Analysis

If an algorithm incorrectly treats an odd resolution (Class III) as unrotated ($\theta_r = 0$), the true normal $\mathbf{n}_k(r)$ is replaced by $\mathbf{n}_k^{(0)}$. For an advective velocity field oriented at polar angle ϕ , the calculated interface flux encounters an angular error $\delta\theta = \alpha$:

$$\tilde{\Phi}_{s,k} = \|\mathbf{j}_s\| \cos(\phi - (k\pi/3 + \alpha))L(r)h \quad (12)$$

The spurious projection residual $\epsilon_k = \cos(\phi - k\pi/3 - \alpha) - \cos(\phi - k\pi/3)$ induces an artificial numerical divergence over the hexagonal cell:

$$\nabla \cdot \mathbf{J}_{\text{err}} = \sum_{k=0}^5 \epsilon_k L(r) h \neq 0 \quad (13)$$

Under non-equilibrium thermodynamics, the entropy generation rate $\dot{S}_{\text{gen},i \rightarrow j}$ across interface k is given by [3, 4]:

$$\dot{S}_{\text{gen},i \rightarrow j} = \Phi_{U,k} \left(\frac{1}{T_j} - \frac{1}{T_i} \right) + \sum_s \Phi_{s,k} \left(\frac{\mu_{s,i}}{T_i} - \frac{\mu_{s,j}}{T_j} \right) \geq 0 \quad (14)$$

Spurious transverse fluxes generated by geometric misalignment disrupt the gradient descent along chemical potential and temperature gradients, artificially producing negative local entropy rates ($\dot{S}_{\text{num}} < 0$). Standardizing the aperture classifier guarantees that $\mathbf{n}_k(r)$ aligns with the underlying manifold geometry, preserving Second-Law compliance across the grid.

3 Implementation Architecture

The TypeScript module `src/spatial/h3_adjacency.ts` incorporates the verified classifier:

```
export type H3ApertureClass = 'CLASS_II' | 'CLASS_III';

export function getApertureClassForResolution(res: number): H3ApertureClass {
  if (!Number.isInteger(res) || res < 0) {
    throw new RangeError(
      'Resolution must be a non-negative integer, received: ${res}'
    );
  }
  return (res % 2 === 0) ? 'CLASS_II' : 'CLASS_III';
}
```

The class `H3AdjacencyGraph` exposes this classifier through instance delegation:

$$\text{graph.getResolutionApertureClass}(r) \equiv \text{getApertureClassForResolution}(r) \quad (15)$$

providing direct access to orientation metadata for spatial thermodynamic monads and finite-volume differential operators.

4 Verification and Invariant Analysis

Table 1 summarizes the discrete aperture classification and edge orientation vectors across representative H3 resolutions.

The test harness confirms that invalid resolution inputs, including negative integers ($r = -1$), non-integers ($r = 1.5$), and non-finite values (NaN , ∞), consistently raise `RangeError`, preserving operational safety and type invariants.

5 Conclusion

The formalization of `getApertureClassForResolution` resolves an essential geometric coupling between recursive DGGs tessellations and spatial physical operators in the *Web of Life* framework. By identifying the alternating $\text{SO}(2)$ rotational phase of Aperture-7 hexagons ($\alpha \approx 19.11^\circ$),

Table 1: Aperture classification, phase rotation, and edge-0 outward normal vectors across H3 resolutions.

Resolution r	Expected Class	Rotation θ_r ($^\circ$)	θ_r (rad)	Normal $\mathbf{n}_0(r)$	First Law $\sum \Delta M$
0	CLASS_II	0.0000 $^\circ$	0.000000	(1.000000, 0.000000)	Conserved (0.0)
1	CLASS_III	19.1063 $^\circ$	0.333473	(0.944911, 0.327327)	Conserved (0.0)
2	CLASS_II	0.0000 $^\circ$	0.000000	(1.000000, 0.000000)	Conserved (0.0)
3	CLASS_III	19.1063 $^\circ$	0.333473	(0.944911, 0.327327)	Conserved (0.0)
4	CLASS_II	0.0000 $^\circ$	0.000000	(1.000000, 0.000000)	Conserved (0.0)
5	CLASS_III	19.1063 $^\circ$	0.333473	(0.944911, 0.327327)	Conserved (0.0)
15	CLASS_III	19.1063 $^\circ$	0.333473	(0.944911, 0.327327)	Conserved (0.0)

thermodynamic monads can project advective and diffusive fluxes across cell interfaces without numerical divergence or unphysical entropy anomalies. This foundation paves the way for higher-order Laplacian diffusion operators, surface hydrology routing, and atmospheric advection kernels on global hexagonal meshes.

References

- [1] J. P. Snyder, “An Equal-Area Map Projection For Polyhedral Globes,” *Cartographica*, vol. 29, no. 1, pp. 10–21, 1992.
- [2] Uber Technologies, “H3: A Hexagonal Hierarchical Spatial Index,” 2018. [Online]. Available: <https://h3geo.org/>
- [3] I. Prigogine, *Introduction to Thermodynamics of Irreversible Processes*, 3rd ed. New York: Interscience Publishers, 1967.
- [4] S. R. de Groot and P. Mazur, *Non-Equilibrium Thermodynamics*. New York: Dover Publications, 1984.
- [5] P. Ranoroarijaona et al., “Web of Life: Planetary-Scale Ecological and Biogeochemical Engine,” 2023. [Online]. Available: <https://github.com/pascalranoroarijaona/WebOfLife>